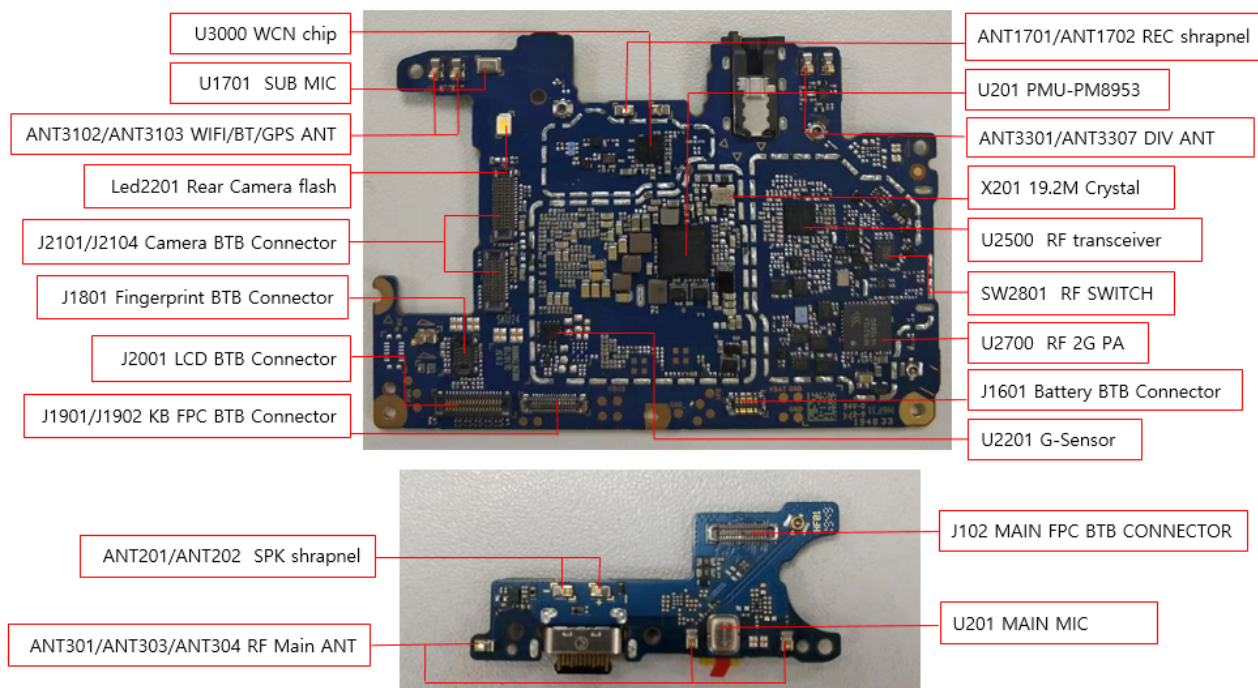


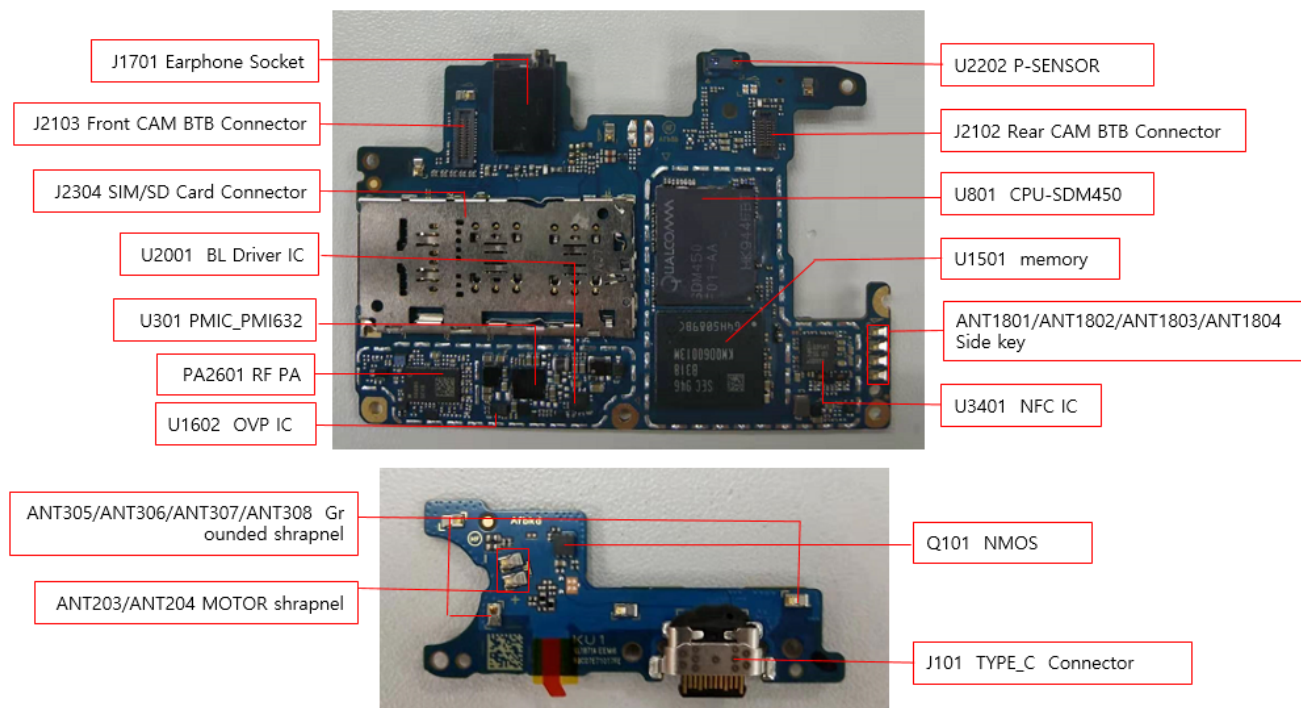
8. Level 3 Repair

8-1. Components Layout

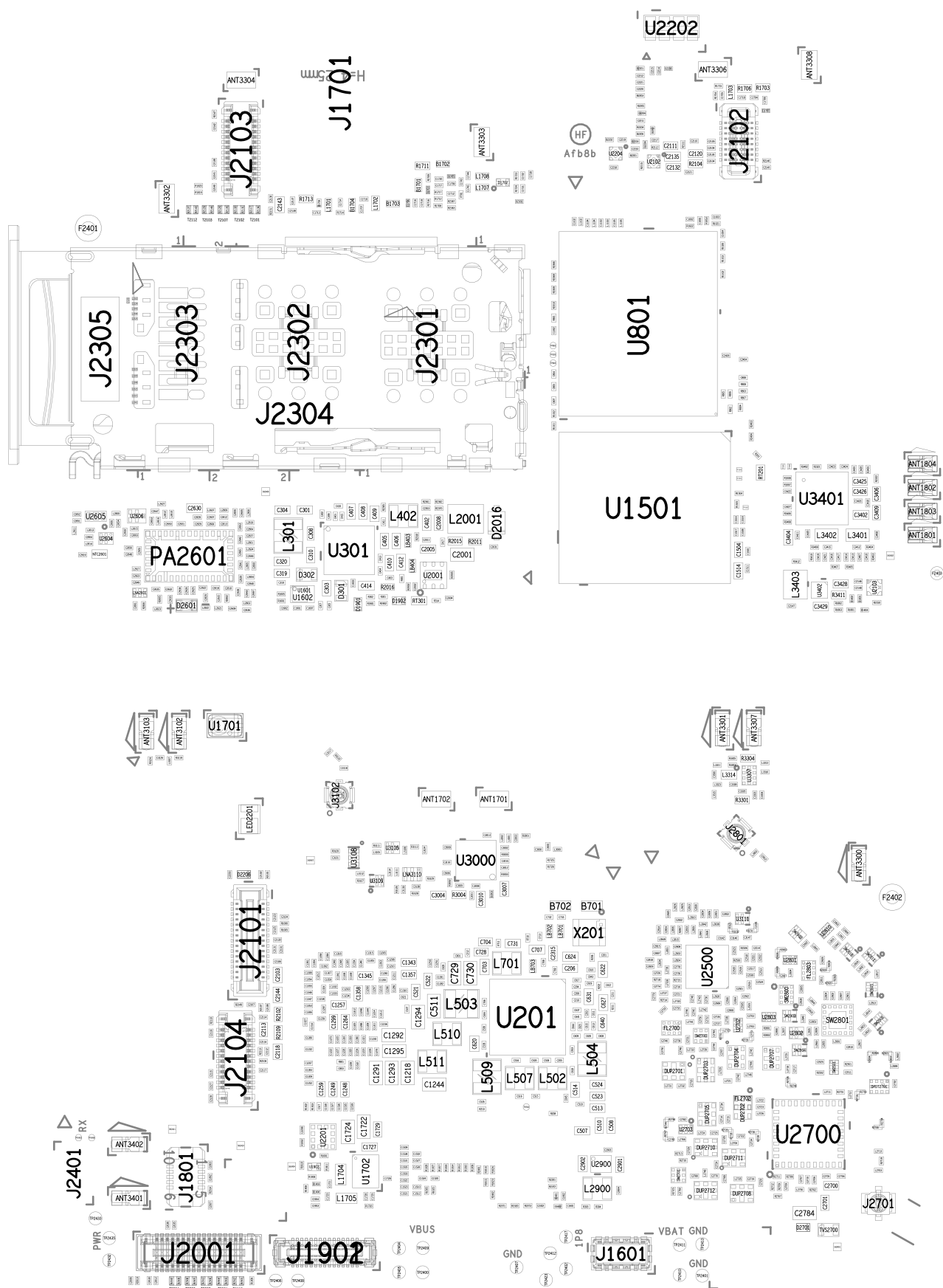
PBA (Bottom)



PRA (TOP)



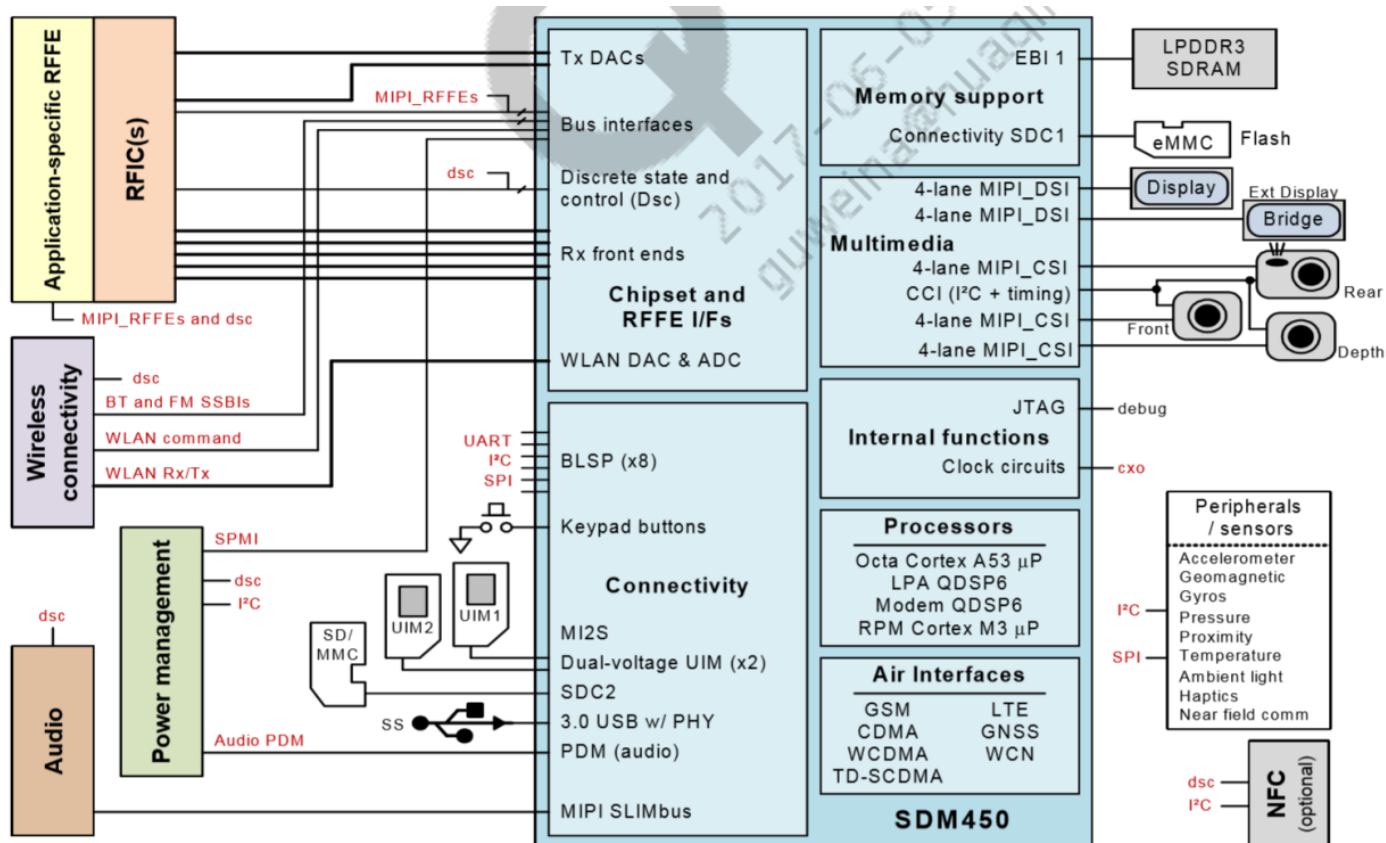
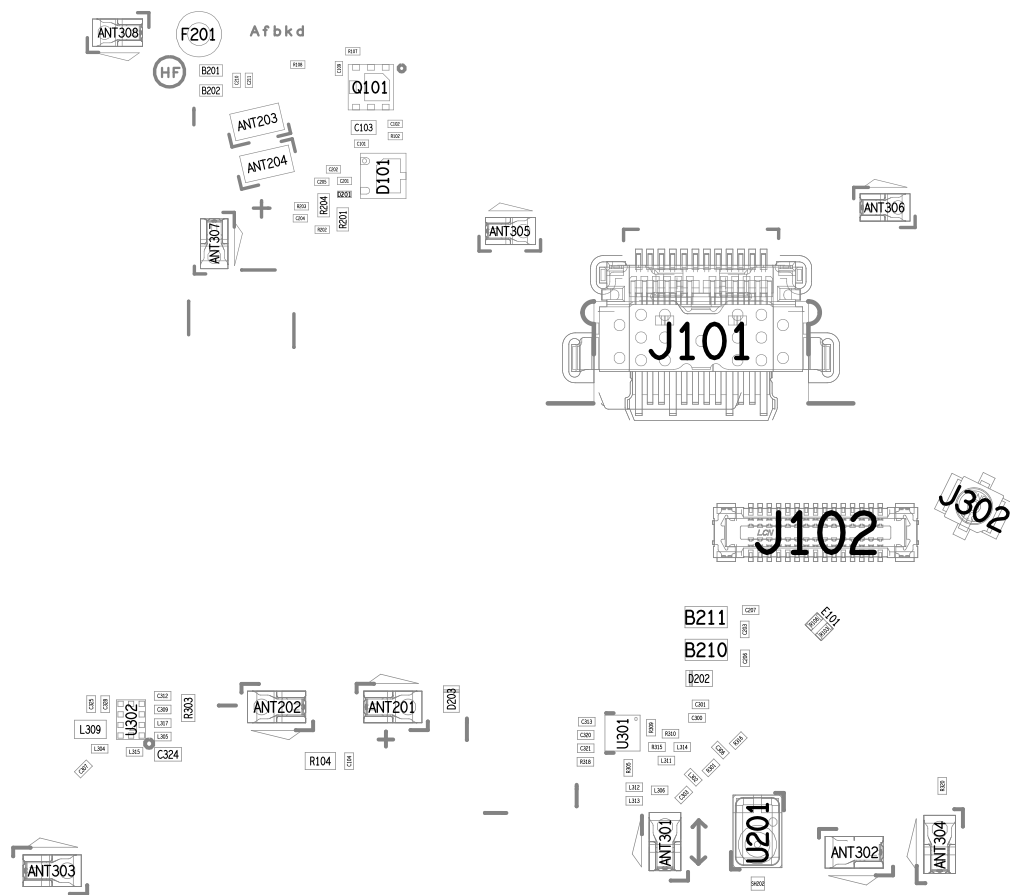
8. Level 3 Repair



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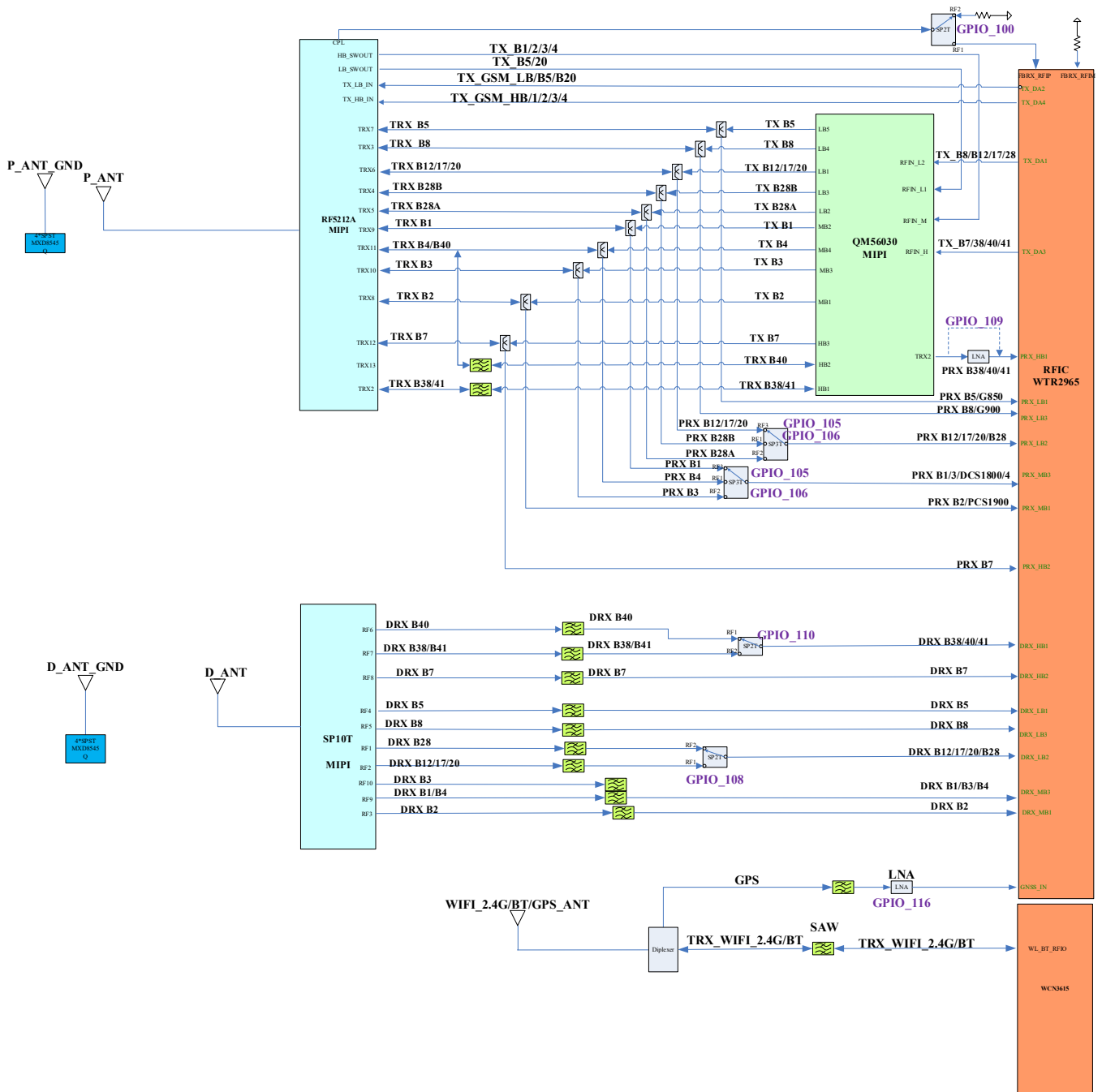
8. Level 3 Repair



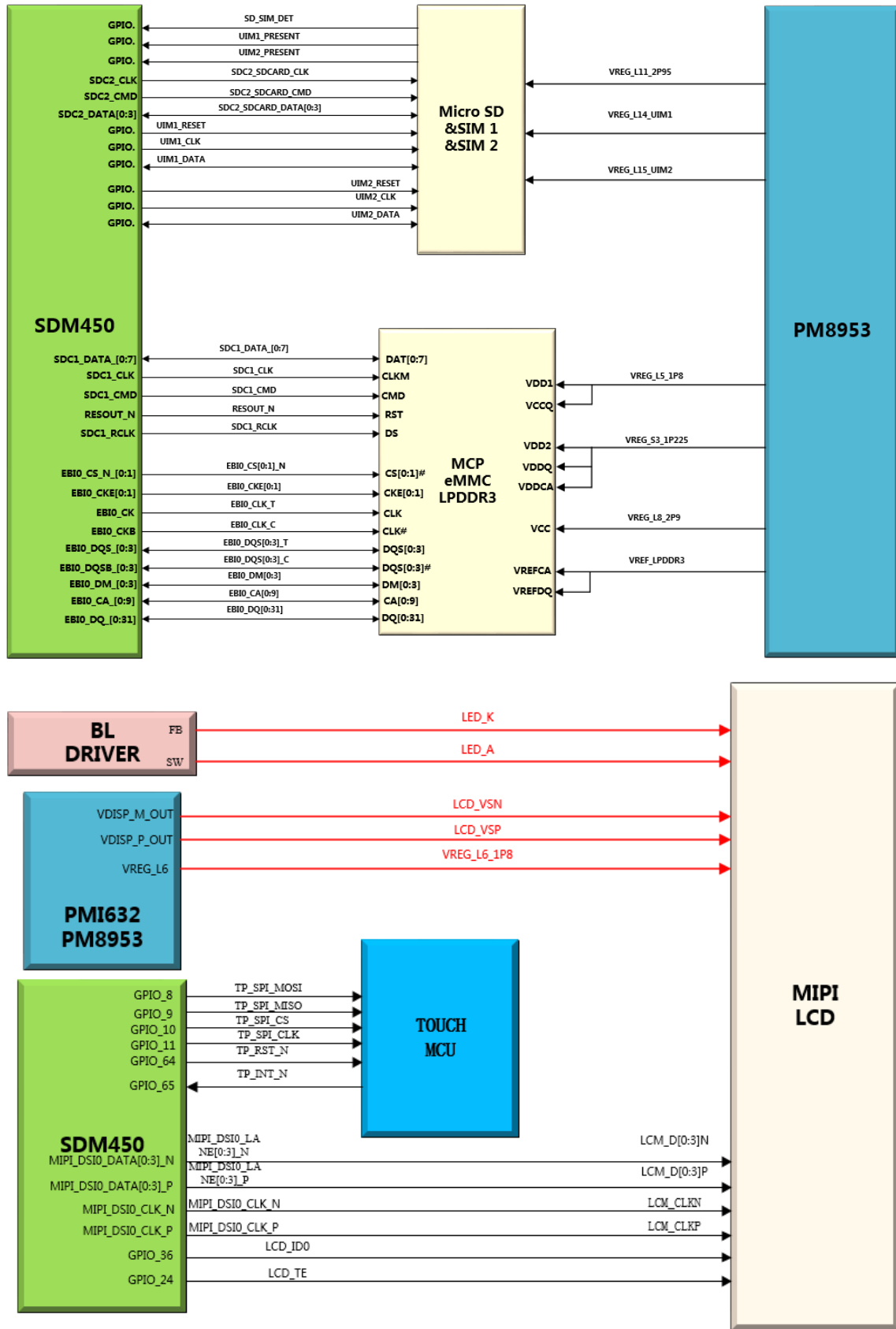
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8. Level 3 Repair



8. Level 3 Repair

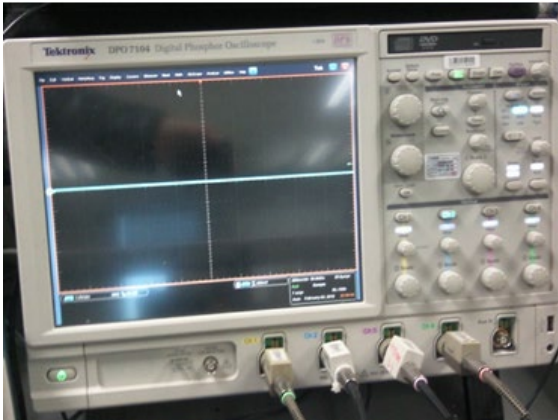


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8. Level 3 Repair

8-3. Flow chart of Troubleshooting.



Oscilloscope



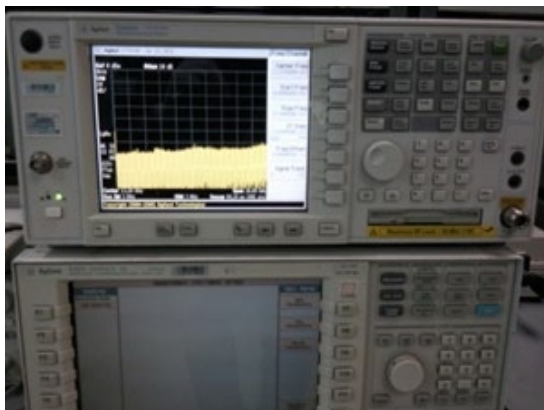
Digital Multimeter



Power Supply



+ driver, ESD Safe Tweezer



8960 & Spectrum Analyzer

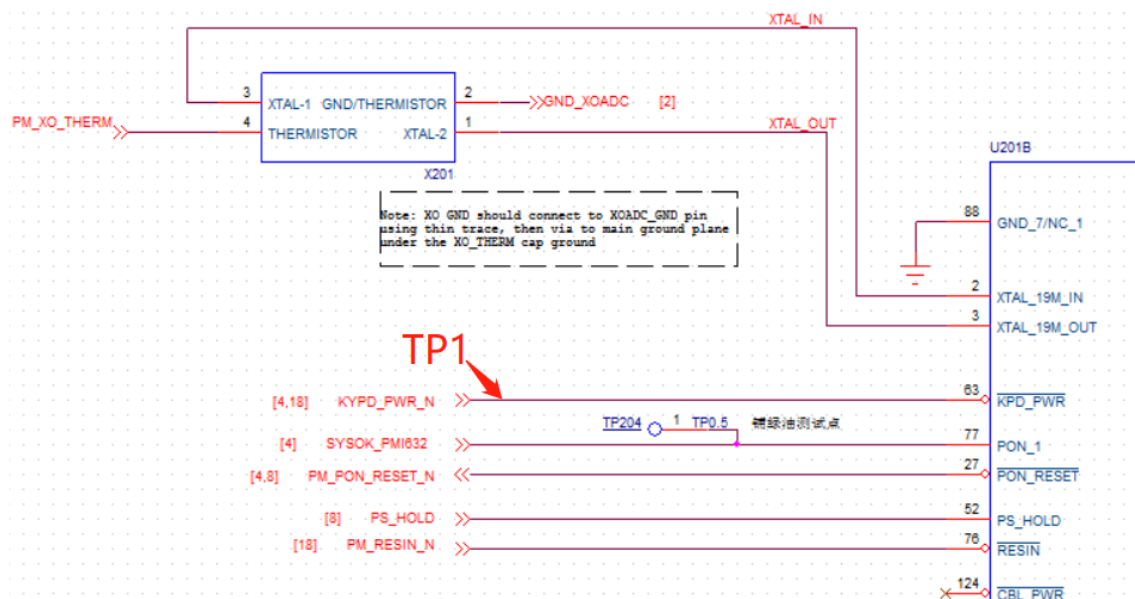
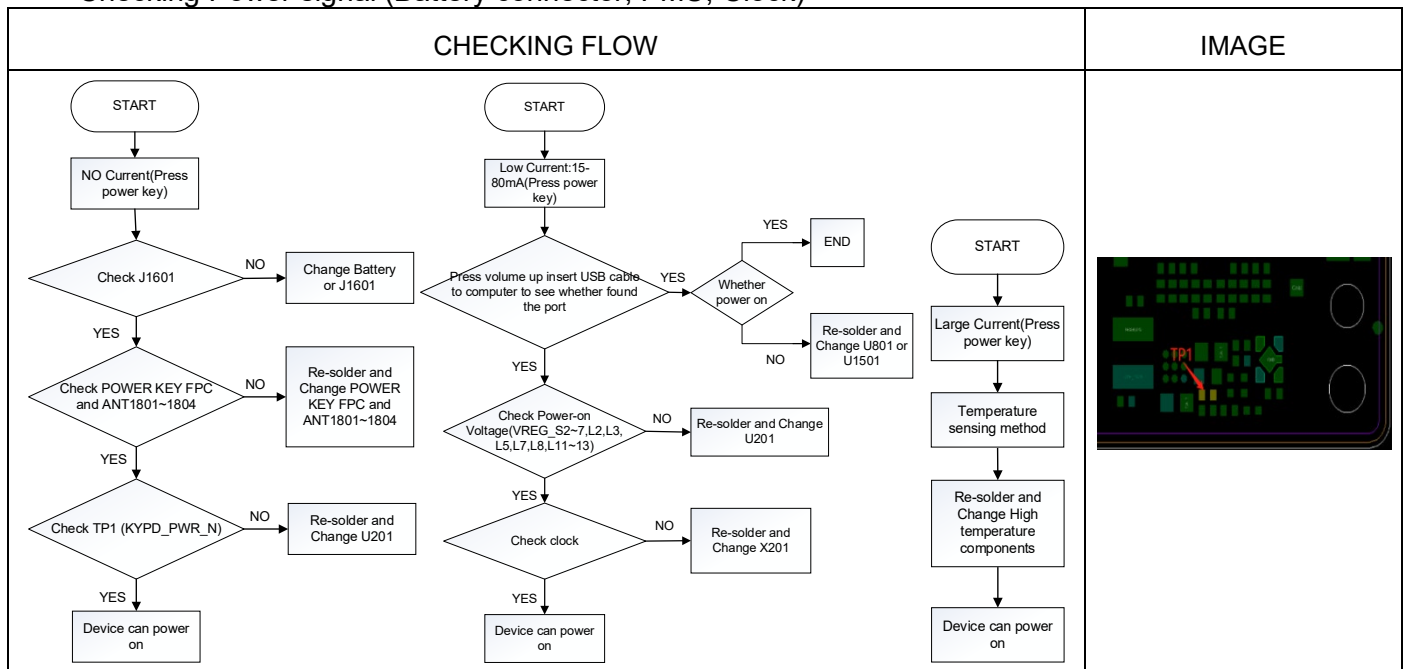


Soldering iron

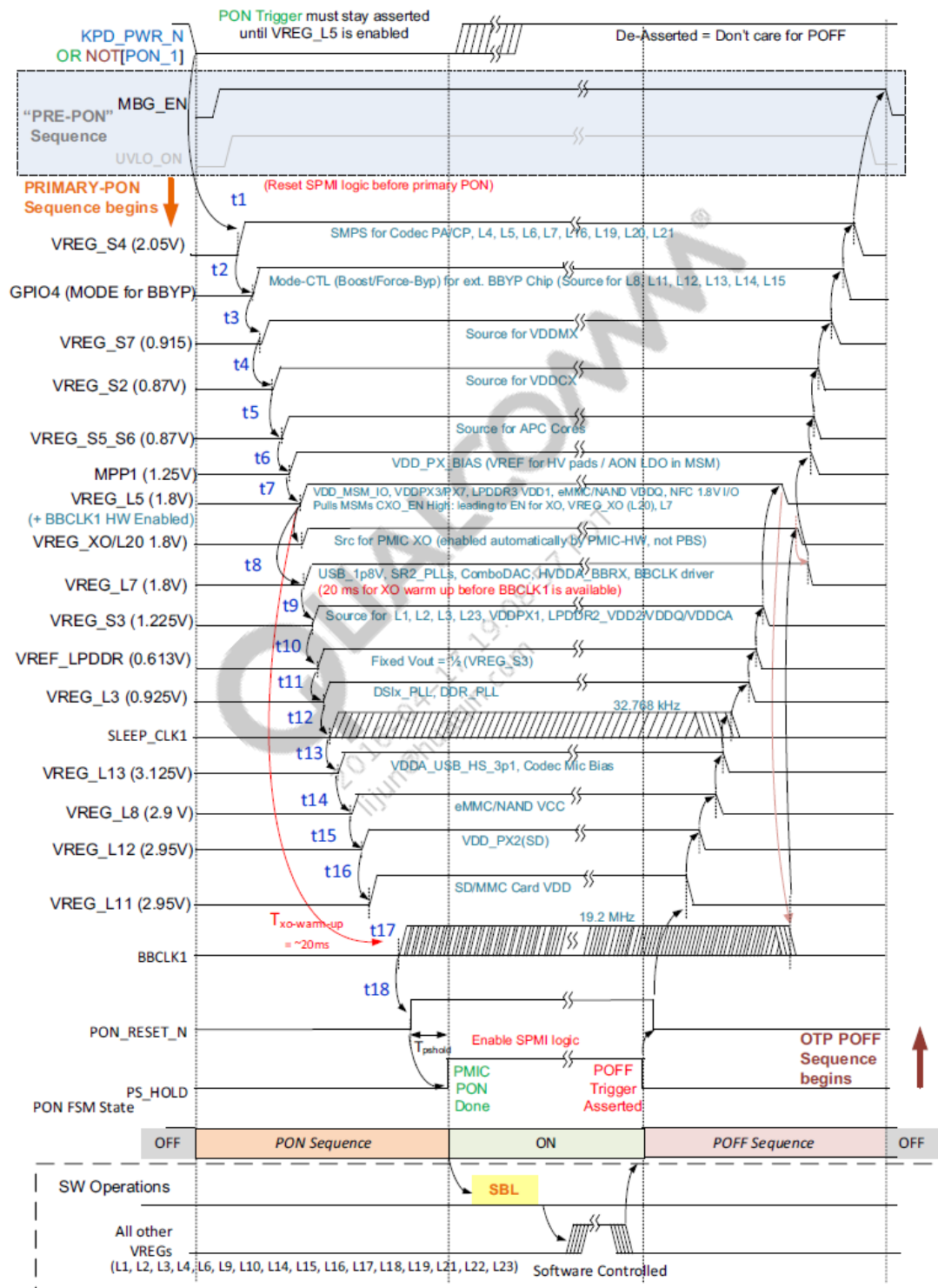
8. Level 3 Repair

8-4-1. Power On

■ Checking Power signal (Battery connector, PMU, Clock)

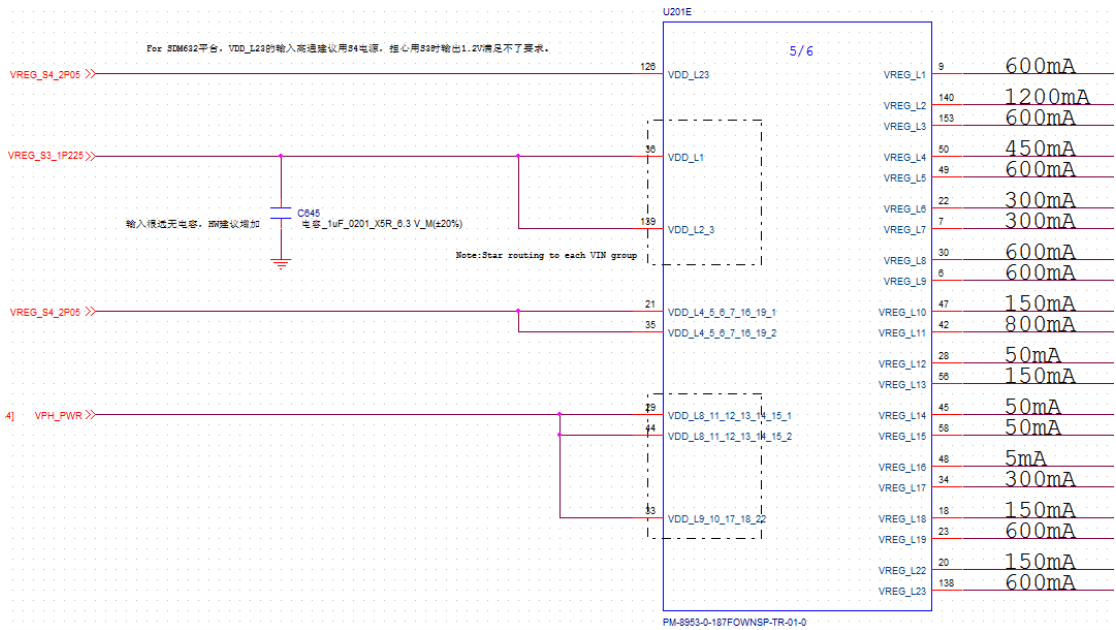


8. Level 3 Repair



Power-on Voltage (VREG_S2~7, L2, L3, L5, L7, L8, L11~13)

8. Level 3 Repair

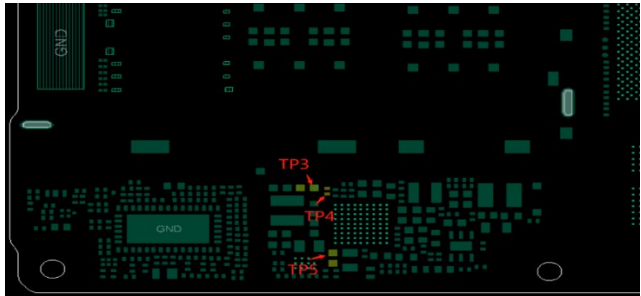
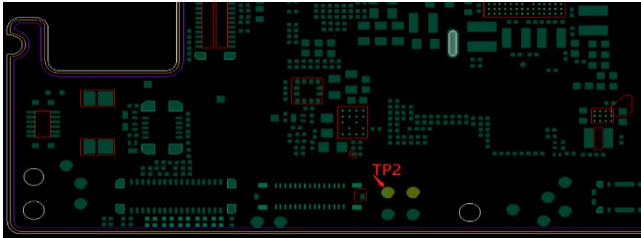
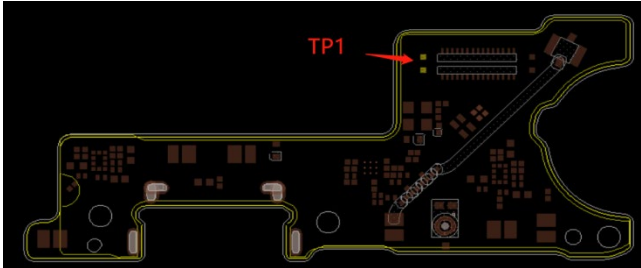


VREG_S2~7, L2, L3, L5, L7, L8, L11~13 schematic diagram

8. Level 3 Repair

8-4-2. Charging

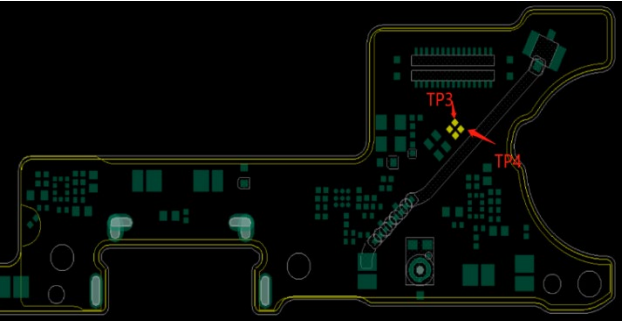
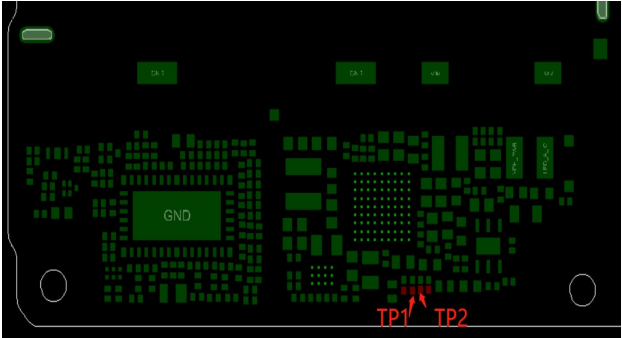
■ The charging controlled by PMU chip PMI632 (U301) and OVP chip U1602

CHECKING FLOW	IMAGE
START Check TP1 NO Re-solder and Change J102 or J101 YES Check VBUS(TP2) NO Re-solder and Change J1901 or FPC YES Check TP3 or TP4 NO Re-solder and Change U1602 YES Check Vbat(TP5) NO Re-solder and Change U301 YES Check battery YES Charge function OK	

8. Level 3 Repair

8-4-3. USB

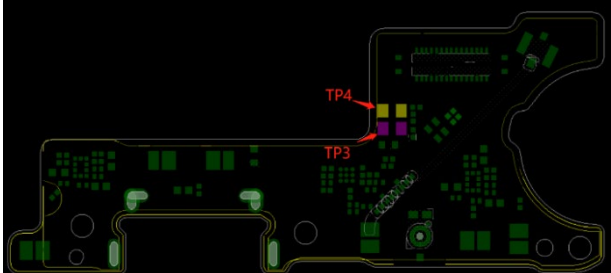
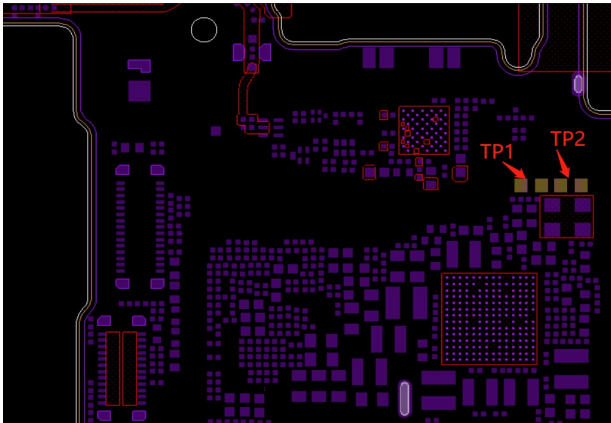
- I/O connector is used as the USB port.

CHECKING FLOW	IMAGE
START Check TP1 and TP2 NO Re-solder and Change U801 or R1901 or R1902 YES Check TP3 and TP4 NO Re-solder and Change J1901 or FPC or J102 or R103 or R106 YES Check J101 soldering NO Re-solder and Change J101 YES USB function OK	

8. Level 3 Repair

8-4-4. Audio speaker

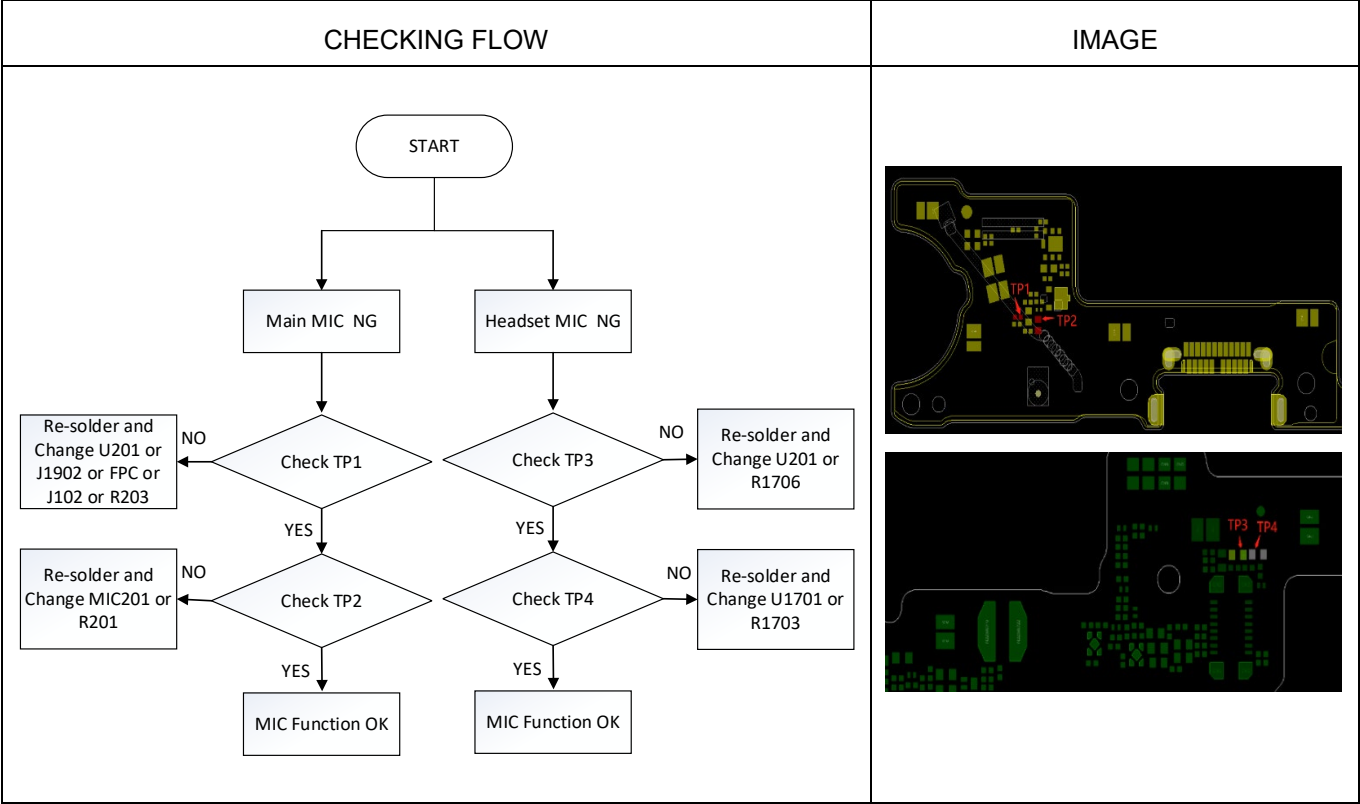
- The Speaker control signals are generated by chip PM8953 (U201), the chip and the speaker are to be checked out.

CHECKING FLOW	IMAGE
START Check TP1-TP2 NO Re-solder and Change U201 or TP1-TP2 YES Check TP3-TP4 NO Re-solder and Change J1902 or J102 or B701/702 or change FPC YES Check ANT201 and ANT202 NO Re-solder and Change ANT201/202 or B210/211 NO Change SPK spring FPC and SPK Speaker function OK	

8. Level 3 Repair

8-4-5. Audio_ MIC

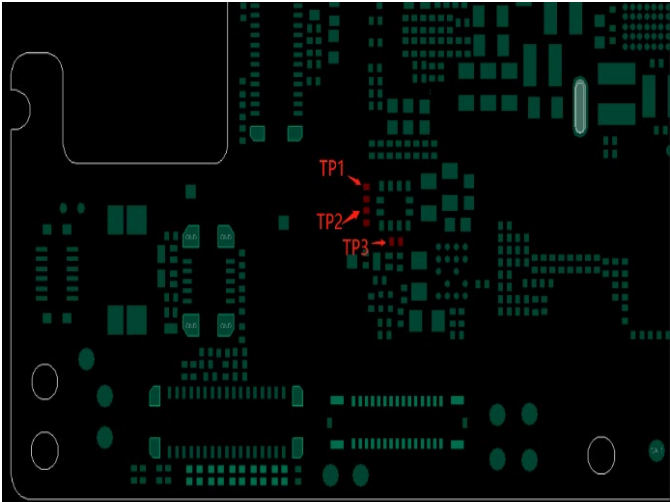
- The MIC control signals are generated by chip PM8953 (U201), the chip and the MIC (main micU201 and headset micU1701) are to be checked out.



8. Level 3 Repair

8-4-6. G sensor

■ The G sensor is calibrated by using SW algorithm.

CHECKING FLOW	IMAGE
START Check VREG_L6_1P8(TP 1) NO Re-solder and Change U201 YES Check TP2-TP3 NO Re-solder and Change R1001 or R1002 YES Re-solder and Change U2201 G sensor function OK	

8. Level 3 Repair

8-4-7. Proximity and light sensor

- Proximity and Light Sensor is worked as below: Control the screen's on/off operation automatically while making phone calls, and adjust the screen brightness according to ambient light.

CHECKING FLOW	IMAGE
START Check VREG_L6_1P8(T P1&TP2) NO → Re-solder and Change R2204 or R2205 or U201 YES → Check VDD_ALPS_3V0(TP3) NO → Re-solder and Change B2203 or B2202 or U201 YES → Check TP4&TP5&TP6 NO → Re-solder and Change B2204 or B2201 or B2206 or U801 YES → Re-solder and Change U2202 ALPS function OK	

8. Level 3 Repair

8-4-11. TOUCH SCREEN

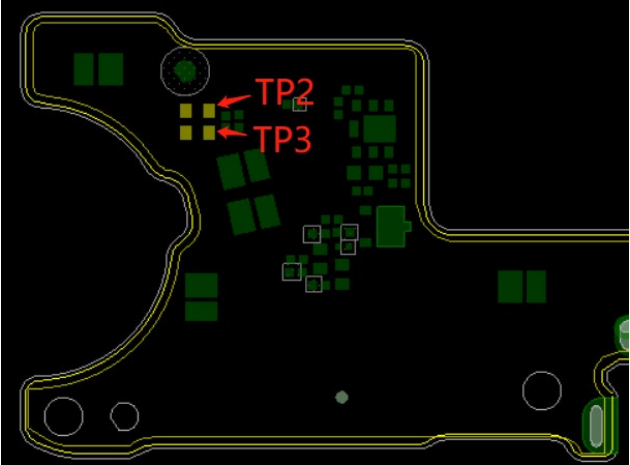
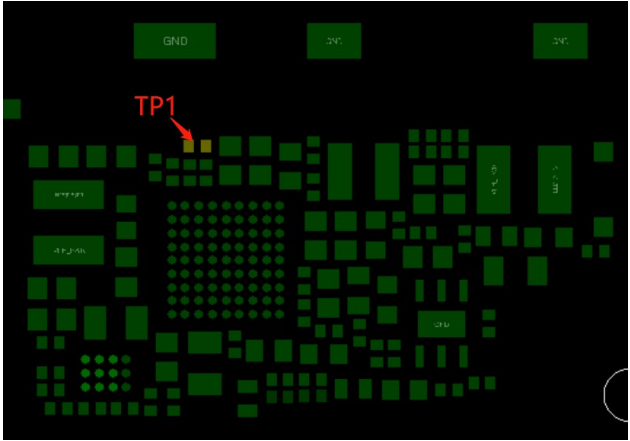
- The Touch control signals are generated by SDM450 (U801). It is assembled with LCD.

CHECKING FLOW	IMAGE
START Check TP1~TP6 NO Re-solder and Change U801 YES Check J2001 NO Re-solder and Change J2001 YES Check TP Module NO Change TP Module YES Touch function OK	

8. Level 3 Repair

8-4-12. Vibrator

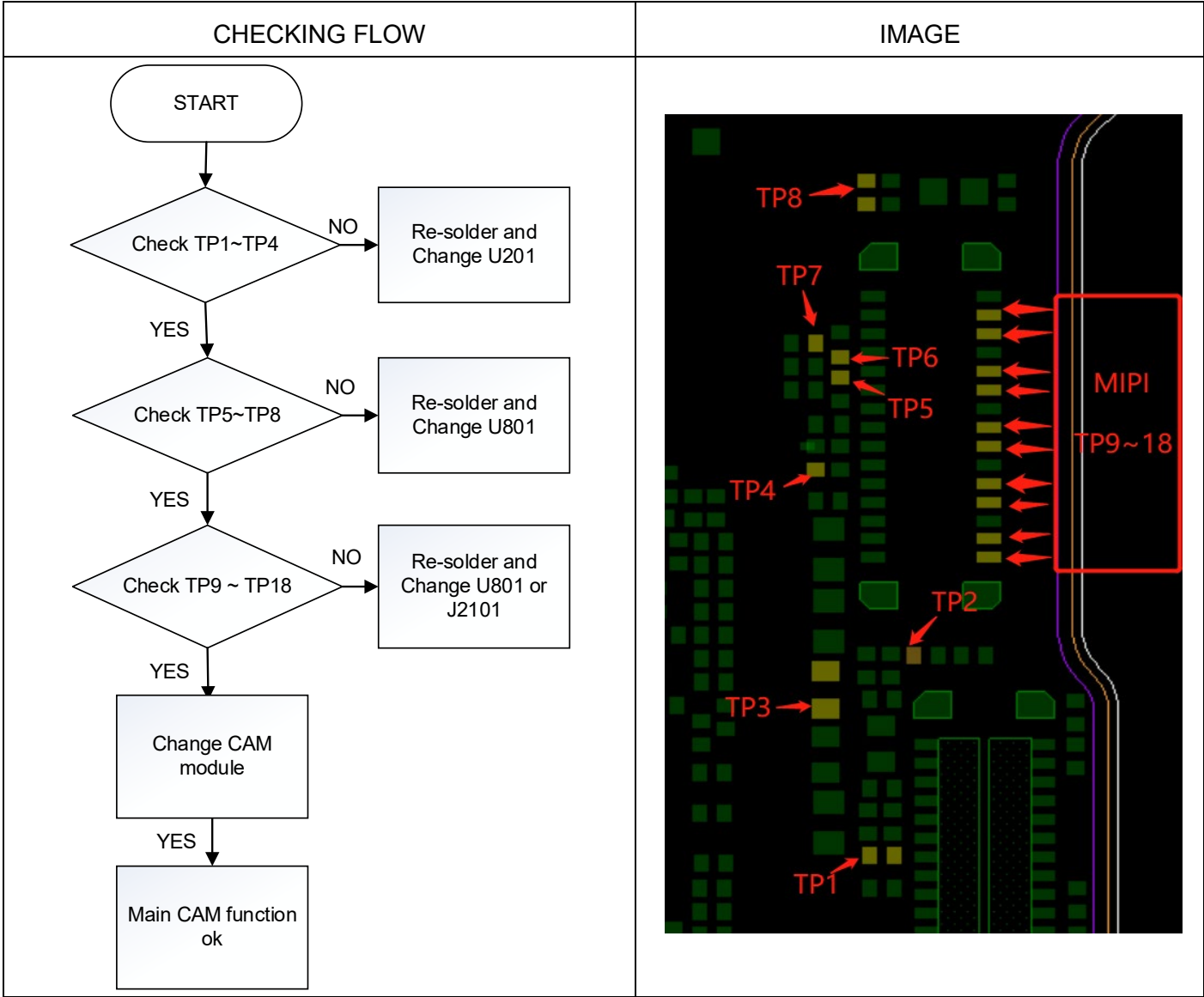
- The Vibrator control signals are generated by PMI632 (U301).

CHECKING FLOW	IMAGE
START Check TP1 NO Re-solder and Change U301 YES Check J102/ J1901/FPC NO Re-solder and Change J102/ J1901/FPC YES Check TP2,TP3 NO Re-solder and Change B201 or B202 YES Check ANT203 ANT204 and vibrator NO Re-solder and Change ANT203 ANT204 or vibrator YES Vibrator function ok	

8. Level 3 Repair

8-4-13.Main Camera

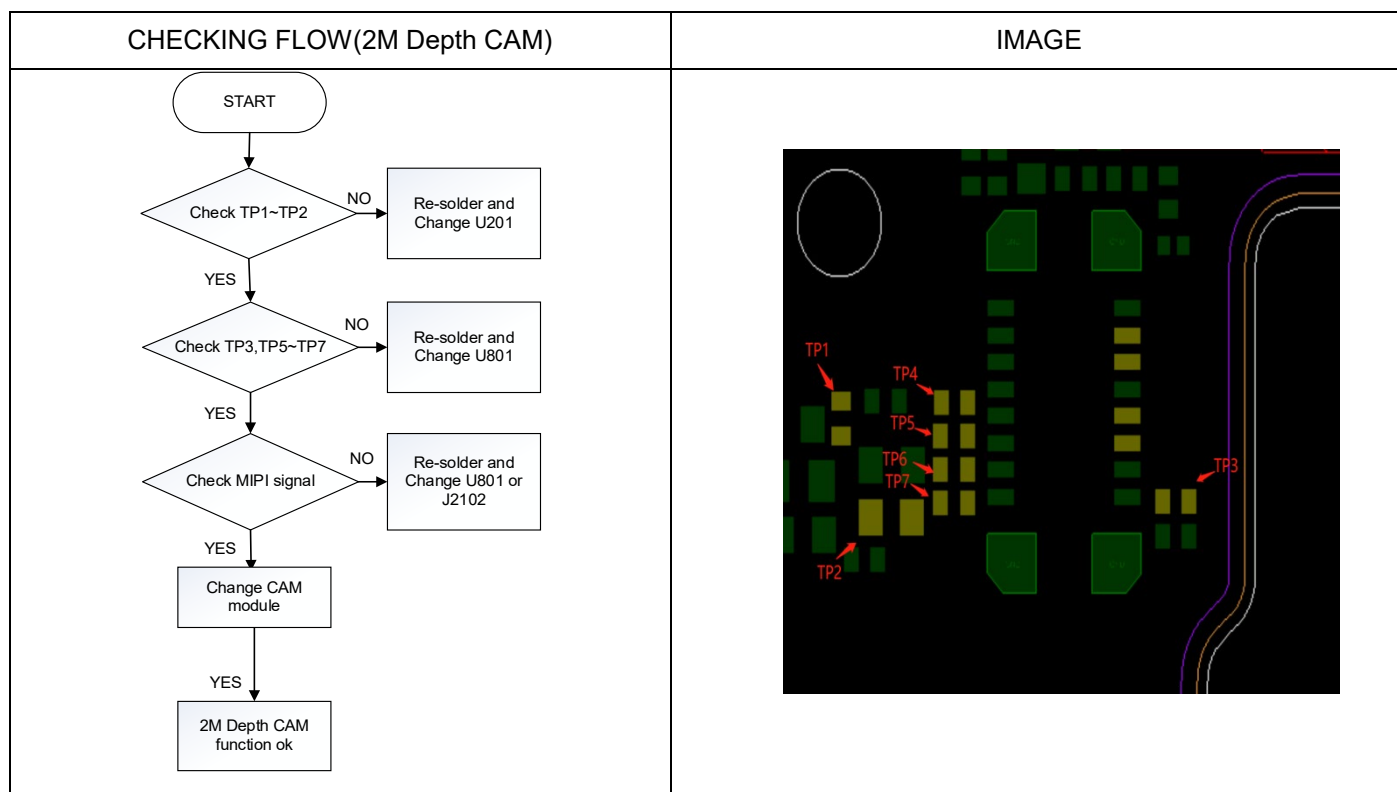
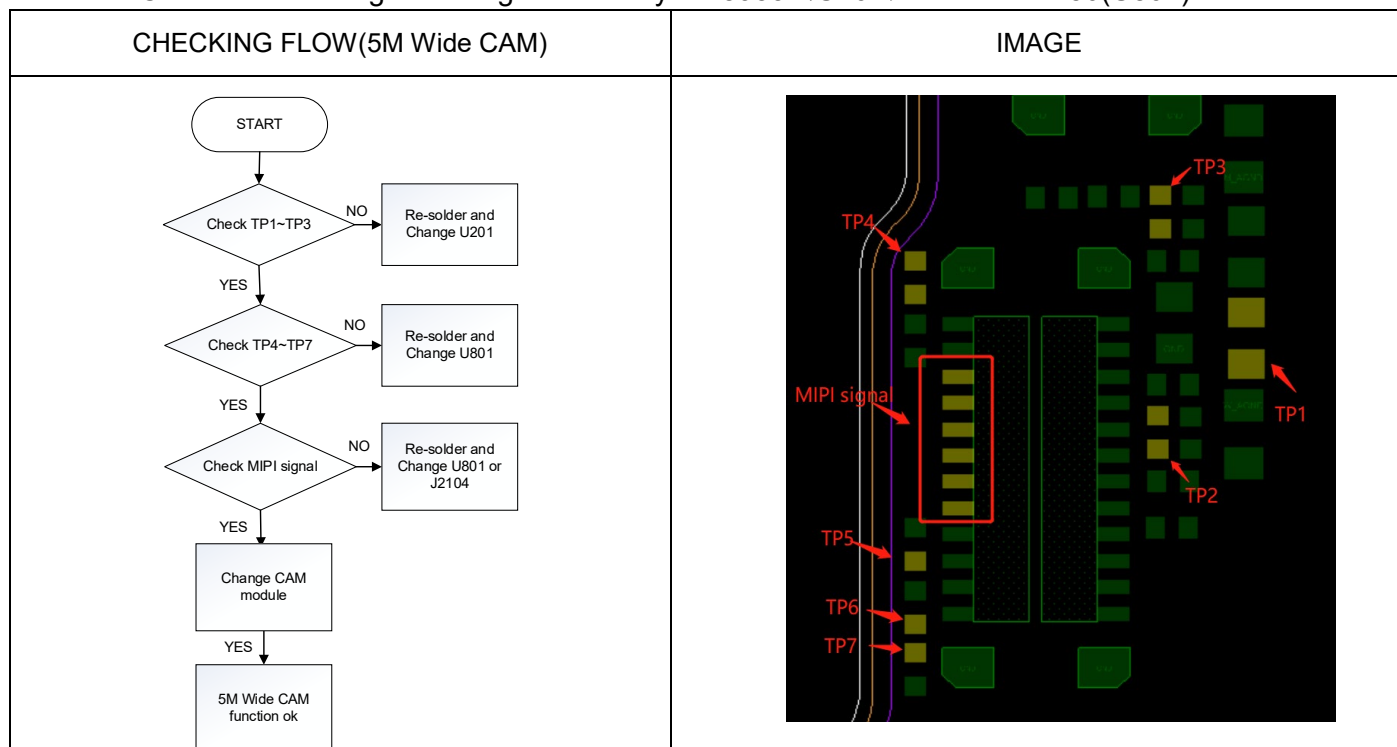
■ The Camera control signals are generated by PM8953 (U201) and SDM450(U801).



8. Level 3 Repair

8-4-14. Rear auxiliary Camera

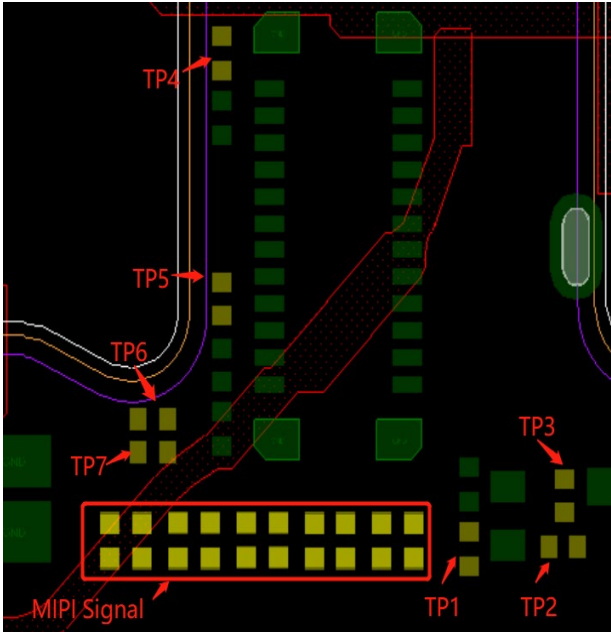
- The Camera control signals are generated by PM8953 (U201) and SDM450(U801).



8. Level 3 Repair

8-4-15. Front Camera

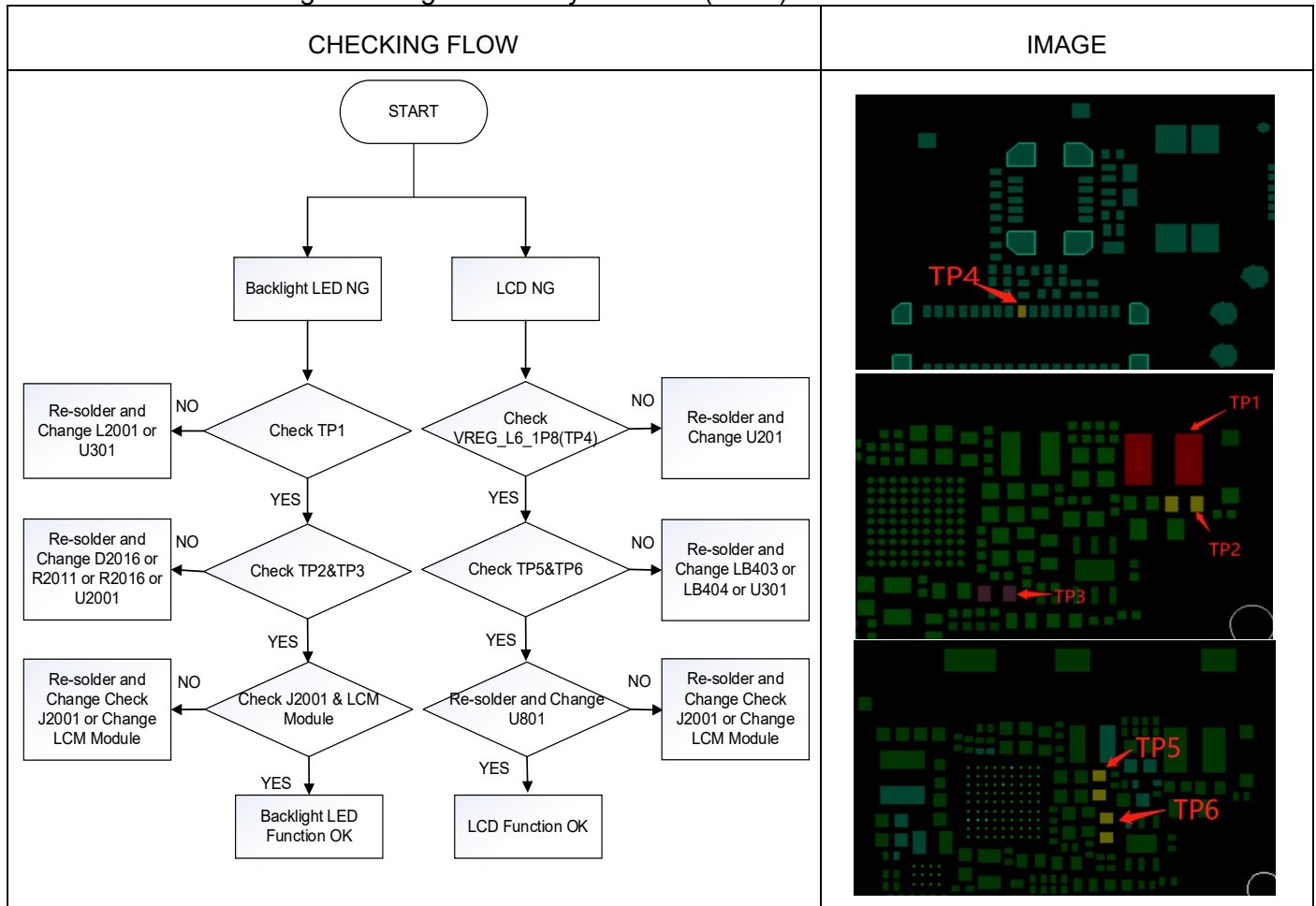
- The Camera control signals are generated by PM8953 (U201) and SDM450(U801).

CHECKING FLOW(Front CAM)	IMAGE
START Check TP1~TP3 NO → Re-solder and Change U201 YES → Check TP4~TP7 NO → Re-solder and Change U801 YES → Check MIPI signal NO → Re-solder and Change U801 or J2103 YES → Change CAM module YES → Front CAM function ok	 The image shows a microscopic view of the front camera module. A red rectangle highlights the MIPI signal line. Red arrows point to test points TP1, TP2, TP3, TP4, TP5, TP6, and TP7. The MIPI signal line is labeled 'MIPI Signal'.

8. Level 3 Repair

8-4-16. LCD

- The LCD control signals are generated by SDM450 (U801).



8-5. Service Schematics

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
A	GRID_0A_1	GRID_0A_2	GRID_0A_3	GRID_0A_4	GRID_0A_5	GRID_0A_6	GRID_0A_7	GRID_0A_8	GRID_0A_9	GRID_0A_10	GRID_0A_11	GRID_0A_12	GRID_0A_13	GRID_0A_14	GRID_0A_15	GRID_0A_16	GRID_0A_17	GRID_0A_18	GRID_0A_19	GRID_0A_20	GRID_0A_21	GRID_0A_22	GRID_0A_23	GRID_0A_24	GRID_0A_25	GRID_0A_26	GRID_0A_27	GRID_0A_28	GRID_0A_29	GRID_0A_30	GRID_0A_31	GRID_0A_32	GRID_0A_33	GRID_0A_34	GRID_0A_35	GRID_0A_36	GRID_0A_37	GRID_0A_38	GRID_0A_39	GRID_0A_40	GRID_0A_41
B	GRID_0B_1	GRID_0B_2	GRID_0B_3	GRID_0B_4	GRID_0B_5	GRID_0B_6	GRID_0B_7	GRID_0B_8	GRID_0B_9	GRID_0B_10	GRID_0B_11	GRID_0B_12	GRID_0B_13	GRID_0B_14	GRID_0B_15	GRID_0B_16	GRID_0B_17	GRID_0B_18	GRID_0B_19	GRID_0B_20	GRID_0B_21	GRID_0B_22	GRID_0B_23	GRID_0B_24	GRID_0B_25	GRID_0B_26	GRID_0B_27	GRID_0B_28	GRID_0B_29	GRID_0B_30	GRID_0B_31	GRID_0B_32	GRID_0B_33	GRID_0B_34	GRID_0B_35	GRID_0B_36	GRID_0B_37	GRID_0B_38	GRID_0B_39	GRID_0B_40	GRID_0B_41
C	GRID_0C_1	GRID_0C_2	GRID_0C_3	GRID_0C_4	GRID_0C_5	GRID_0C_6	GRID_0C_7	GRID_0C_8	GRID_0C_9	GRID_0C_10	GRID_0C_11	GRID_0C_12	GRID_0C_13	GRID_0C_14	GRID_0C_15	GRID_0C_16	GRID_0C_17	GRID_0C_18	GRID_0C_19	GRID_0C_20	GRID_0C_21	GRID_0C_22	GRID_0C_23	GRID_0C_24	GRID_0C_25	GRID_0C_26	GRID_0C_27	GRID_0C_28	GRID_0C_29	GRID_0C_30	GRID_0C_31	GRID_0C_32	GRID_0C_33	GRID_0C_34	GRID_0C_35	GRID_0C_36	GRID_0C_37	GRID_0C_38	GRID_0C_39	GRID_0C_40	GRID_0C_41
D	GRID_0D_1	GRID_0D_2	GRID_0D_3	GRID_0D_4	GRID_0D_5	GRID_0D_6	GRID_0D_7	GRID_0D_8	GRID_0D_9	GRID_0D_10	GRID_0D_11	GRID_0D_12	GRID_0D_13	GRID_0D_14	GRID_0D_15	GRID_0D_16	GRID_0D_17	GRID_0D_18	GRID_0D_19	GRID_0D_20	GRID_0D_21	GRID_0D_22	GRID_0D_23	GRID_0D_24	GRID_0D_25	GRID_0D_26	GRID_0D_27	GRID_0D_28	GRID_0D_29	GRID_0D_30	GRID_0D_31	GRID_0D_32	GRID_0D_33	GRID_0D_34	GRID_0D_35	GRID_0D_36	GRID_0D_37	GRID_0D_38	GRID_0D_39	GRID_0D_40	GRID_0D_41
E	GRID_0E_1	GRID_0E_2	GRID_0E_3	GRID_0E_4	GRID_0E_5	GRID_0E_6	GRID_0E_7	GRID_0E_8	GRID_0E_9	GRID_0E_10	GRID_0E_11	GRID_0E_12	GRID_0E_13	GRID_0E_14	GRID_0E_15	GRID_0E_16	GRID_0E_17	GRID_0E_18	GRID_0E_19	GRID_0E_20	GRID_0E_21	GRID_0E_22	GRID_0E_23	GRID_0E_24	GRID_0E_25	GRID_0E_26	GRID_0E_27	GRID_0E_28	GRID_0E_29	GRID_0E_30	GRID_0E_31	GRID_0E_32	GRID_0E_33	GRID_0E_34	GRID_0E_35	GRID_0E_36	GRID_0E_37	GRID_0E_38	GRID_0E_39	GRID_0E_40	GRID_0E_41
F	GRID_0F_1	GRID_0F_2	GRID_0F_3	GRID_0F_4	GRID_0F_5	GRID_0F_6	GRID_0F_7	GRID_0F_8	GRID_0F_9	GRID_0F_10	GRID_0F_11	GRID_0F_12	GRID_0F_13	GRID_0F_14	GRID_0F_15	GRID_0F_16	GRID_0F_17	GRID_0F_18	GRID_0F_19	GRID_0F_20	GRID_0F_21	GRID_0F_22	GRID_0F_23	GRID_0F_24	GRID_0F_25	GRID_0F_26	GRID_0F_27	GRID_0F_28	GRID_0F_29	GRID_0F_30	GRID_0F_31	GRID_0F_32	GRID_0F_33	GRID_0F_34	GRID_0F_35	GRID_0F_36	GRID_0F_37	GRID_0F_38	GRID_0F_39	GRID_0F_40	GRID_0F_41
G	GRID_0G_1	GRID_0G_2	GRID_0G_3	GRID_0G_4	GRID_0G_5	GRID_0G_6	GRID_0G_7	GRID_0G_8	GRID_0G_9	GRID_0G_10	GRID_0G_11	GRID_0G_12	GRID_0G_13	GRID_0G_14	GRID_0G_15	GRID_0G_16	GRID_0G_17	GRID_0G_18	GRID_0G_19	GRID_0G_20	GRID_0G_21	GRID_0G_22	GRID_0G_23	GRID_0G_24	GRID_0G_25	GRID_0G_26	GRID_0G_27	GRID_0G_28	GRID_0G_29	GRID_0G_30	GRID_0G_31	GRID_0G_32	GRID_0G_33	GRID_0G_34	GRID_0G_35	GRID_0G_36	GRID_0G_37	GRID_0G_38	GRID_0G_39	GRID_0G_40	GRID_0G_41
H	GRID_0H_1	GRID_0H_2	GRID_0H_3	GRID_0H_4	GRID_0H_5	GRID_0H_6	GRID_0H_7	GRID_0H_8	GRID_0H_9	GRID_0H_10	GRID_0H_11	GRID_0H_12	GRID_0H_13	GRID_0H_14	GRID_0H_15	GRID_0H_16	GRID_0H_17	GRID_0H_18	GRID_0H_19	GRID_0H_20	GRID_0H_21	GRID_0H_22	GRID_0H_23	GRID_0H_24	GRID_0H_25	GRID_0H_26	GRID_0H_27	GRID_0H_28	GRID_0H_29	GRID_0H_30	GRID_0H										